Sampling Distributions & Central Limit Theorem

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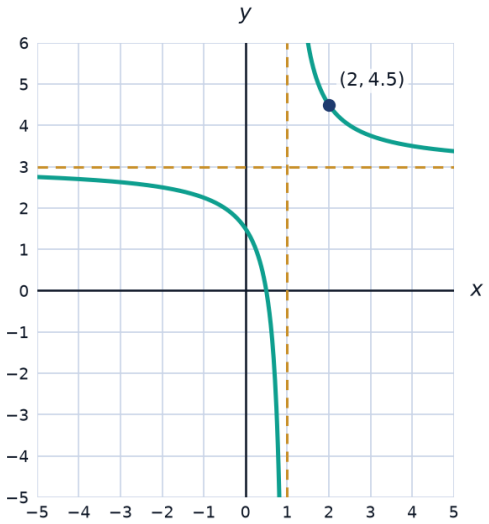
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: What is a sampling distribution?**

Explain the difference between a population distribution and the sampling distribution of a sample mean $\bar{x}$.

2 **[Foundational] • Target: What is a sampling distribution?**

Explain why the sampling distribution of $\bar{x}$ has less variability than the original population distribution.



3 **[Foundational] • Target: Mean and standard deviation of the sampling distribution**

A population has mean $\mu = 60$ and standard deviation $\sigma = 12$. For samples of size $n = 36$, find the mean and standard deviation of the sampling distribution of $\bar{x}$.

4 **[Foundational] • Target: Mean and standard deviation of the sampling distribution**

Using the same population, find the standard error for a larger sample of size $n = 144$, and explain what happens to the standard error as sample size increases.

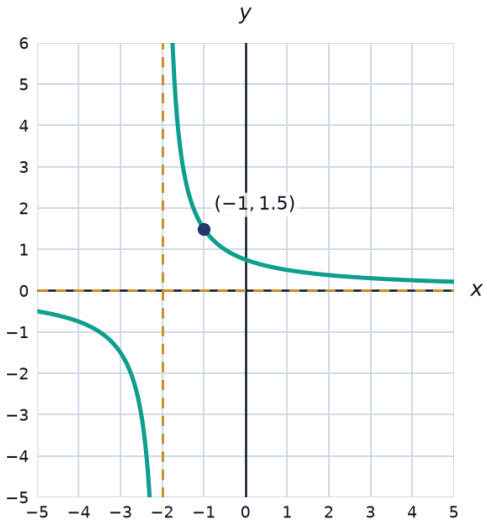
5 **[Foundational] • Target: The Central Limit Theorem**

State the Central Limit Theorem in your own words, including what happens to the shape of the sampling distribution of $\bar{x}$ as sample size n increases.

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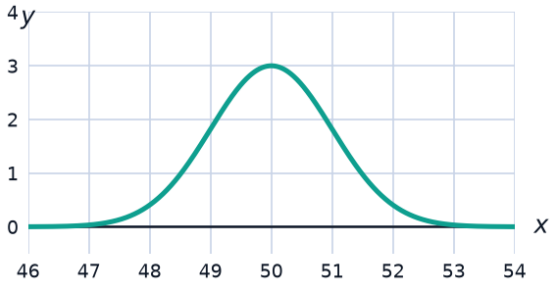
6 [Foundational] • Target: The Central Limit Theorem

Explain why the Central Limit Theorem is considered one of the most important results in statistics, referencing its practical usefulness.



7 [Foundational] • Target: Applying the Central Limit Theorem

A population of package weights has mean $\mu = 50$ lbs and standard deviation $\sigma = 8$ lbs (population shape unknown). For a random sample of $n = 64$ packages, find the probability the sample mean weight exceeds 52 lbs, given $P(Z > 2) \approx 0.0228$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sampling distribution of a sample proportion

A population has a true proportion $p = 0.4$ favoring a proposal. For samples of size $n = 100$, find the mean and standard deviation of the sampling distribution of the sample proportion $\hat{p}$, using $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$.

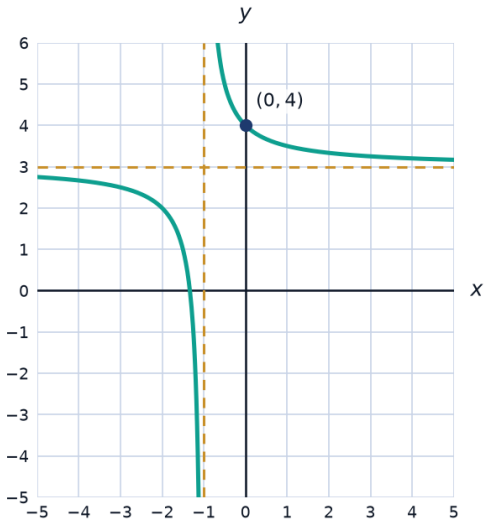
9 [Intermediate] • Target: Sampling distribution of a sample proportion

State the two conditions typically required to treat the sampling distribution of $\hat{p}$ as approximately normal.

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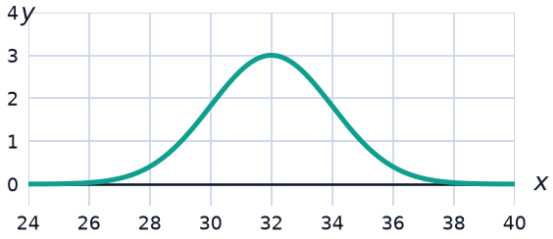
10 **[Intermediate]** • **Target: Comparing sampling variability across sample sizes**

Explain, in the context of opinion polling, why a poll of 1000 people gives a more reliable estimate of a population proportion than a poll of 50 people, connecting your answer to the formula for $\sigma_{\hat{p}}$.



11 **[Intermediate]** • **Target: Applying the sampling distribution of the sample mean**

A population of commute times has mean $\mu = 32$ minutes and standard deviation $\sigma = 10$ minutes. For a random sample of $n = 25$ commuters, find the probability the sample mean commute time exceeds 35 minutes, given $P(Z > 1.5) \approx 0.0668$.



12 **[Intermediate]** • **Target: Sample size and the CLT threshold**

A population of insurance claim amounts is strongly right-skewed. Explain why a sample of size $n = 15$ might not be large enough to safely use CLT-based normal methods, but a sample of size $n = 200$ likely would be.

13 **[Intermediate]** • **Target: Standard error vs. standard deviation**

Explain the difference between the standard deviation of an individual population value and the standard error of a sample mean $\bar{x}$.

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14 **[Intermediate]** • **Target: Standard error vs. standard deviation**

A quality control engineer at a bottling plant knows that individual bottle fill volumes are approximately normal with mean 500 mL and standard deviation 8 mL. Every hour, the engineer measures a sample of 16 bottles and computes the sample mean fill volume. Describe the shape, mean, and standard deviation of the sampling distribution of this hourly sample mean, and explain whether the Central Limit Theorem's usual $n \geq 30$ guideline is even needed to justify treating this sampling distribution as normal.

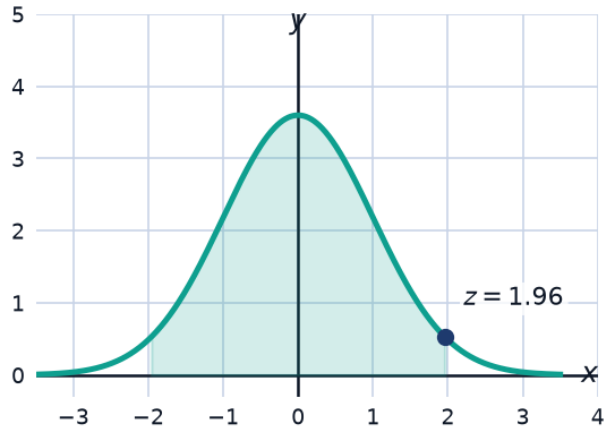
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Standard error vs. standard deviation** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

A political pollster samples 900 voters and calculates a sample proportion $\hat{p} = 0.53$ supporting a ballot measure, from a true population proportion of $p = 0.50$. Verify that both conditions ($np \geq 10$ and $n(1-p) \geq 10$) are satisfied for treating the sampling distribution of $\hat{p}$ as approximately normal, then find the standard deviation of this sampling distribution, to four decimal places.

16 **[Challenging]** • **Target: Standard error vs. standard deviation** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

A large factory produces light bulbs whose individual lifespans are strongly right-skewed (most bulbs fail early, but a few last an unusually long time), with population mean $\mu = 800$ hours and standard deviation $\sigma = 250$ hours. An engineer takes a random sample of $n = 64$ bulbs and computes the sample mean lifespan. Explain, step by step, why it is reasonable to treat this sampling distribution of $\bar{x}$ as approximately normal despite the population itself being skewed, then compute the standard error of $\bar{x}$.



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17 [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 2.02 Experimental Design & Scope of Inference

An online retailer knows that 22% of all website visitors make a purchase. On a particular day, the retailer records the browsing sessions of a random sample of 500 visitors and calculates the sample proportion who made a purchase. Verify the two conditions needed to treat this sampling distribution as approximately normal, then compute the standard deviation of the sampling distribution of the sample proportion, to four decimal places, and describe what this standard deviation represents in the context of comparing many different days' worth of visitor samples.

18 [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

Level 2 · MediumExplain the difference between a single-blind experiment and a double-blind experiment, and describe why a placebo is used in medical trials.

19 [Challenging] • Target: What is a sampling distribution? • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

Level 3 · AdvancedTen athletes participate in a vertical jump training program. Their jump heights (in inches) before and after the program are recorded. To test whether the program increased jump height, explain why a paired t-test is appropriate rather than a two-sample t-test, state the null and alternative hypotheses, and specify the degrees of freedom.

20 [Challenging] • Target: Mean and standard deviation of the sampling distribution • Cross-Lesson: 3.03 The Normal Distribution

Level 2 · MediumFind the critical value z^* required to construct a 98% confidence interval for a population proportion, rounded to three decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: What is a sampling distribution?

A population has mean $\mu = 100$ and standard deviation $\sigma = 60$. For samples of size $n = 100$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error), using $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$.

a. 6

b. 5

c. 7

d. 60

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22 [Foundational] • Target: Mean and standard deviation of the sampling distribution

A population has a true proportion $p = 0.40$. For samples of size $n = 200$, find the standard deviation of the sampling distribution of the sample proportion $\hat{p}$, to four decimal places, using $\sigma_{\hat{p}} = \sqrt{p(1-p)/n}$.

a. 0.0346

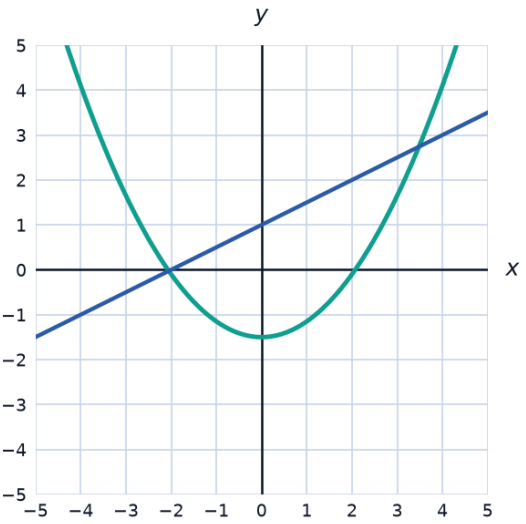
b. 0.0246

c. 0.2400

d. 0.0446

23 [Foundational] • Target: The Central Limit Theorem

A population has mean $\mu = 143$ and standard deviation $\sigma = 90$ (population shape unknown). For a random sample of $n = 36$, find the probability the sample mean is above 158, using the Central Limit Theorem.



a. 11.87%

b. 15.87%

c. 84.13%

d. 19.87%

24 [Foundational] • Target: Applying the Central Limit Theorem

A population has mean $\mu = 48$ and standard deviation $\sigma = 132$ (population shape unknown). For a random sample of $n = 144$, find the probability the sample mean is below 81, using the Central Limit Theorem.

a. 96.87%

b. 95.87%

c. 0.13%

d. 99.87%

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25 [Foundational] • Target: Sampling distribution of a sample proportion

A pollster samples $n = 500$ voters, from a population with true proportion $p = 0.20$ supporting a measure. Verify that both conditions ($np \geq 10$ and $n(1 - p) \geq 10$) are satisfied, then find the standard deviation of the sampling distribution of $\hat{p}$, to four decimal places.

a. 0.0259

b. 0.2000

c. 0.0179

d. 0.0099

26 [Foundational] • Target: Comparing sampling variability across sample sizes

A population has standard deviation $\sigma = 160$. Find the sample size n needed so that the standard error $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$ equals 20.

a. 72

b. 64

c. 56

d. 8

27 [Foundational] • Target: Applying the sampling distribution of the sample mean

A researcher is studying the average height of adult males in a city. They collected a sample of 100 adult males and found the sample mean height to be 175 cm with a sample standard deviation of 8 cm. If the margin of error for a 95% confidence interval is calculated to be approximately 1.57 cm, what is the minimum sample size required to achieve a margin of error of at most 1 cm, assuming the sample standard deviation remains the same?

a. 154

b. 246

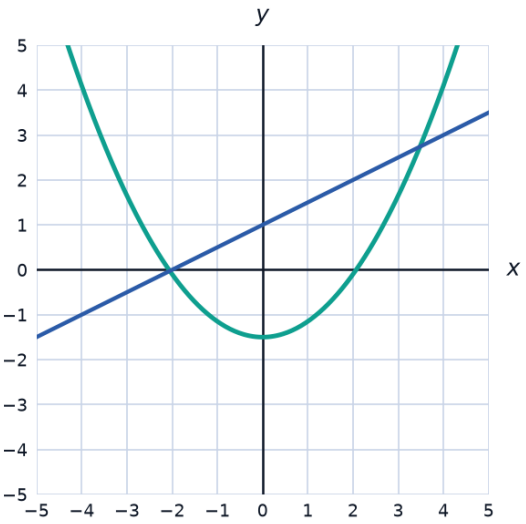
c. 197

d. 97

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28 [Foundational] • Target: Sample size and the CLT threshold

A sample of size $n = 64$ has a sample mean of 80 and standard deviation 16. What is the standard error of the mean?



a. 0.25

b. 1.0

c. 2.0

d. 4.0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Standard error vs. standard deviation

A researcher wants a 95% confidence interval for a proportion with margin of error at most 0.043, using a planning value $p^* = 0.4$. What sample size n is required ($z^* = 1.96$)?

a. 529

b. 499

c. 484

d. 469

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30 [Intermediate] • Target: What is a sampling distribution?

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

31 [Intermediate] • Target: Mean and standard deviation of the sampling distribution

A quality control process monitors the weight, W grams, of chocolate bars produced by a machine. A random sample of 50 chocolate bars was taken, and their weights were recorded. The sample mean weight was 102.5 grams, and the sample standard deviation was 3.5 grams. A) Calculate the standard error of the mean. B) Construct a 95% confidence interval for the mean weight of the chocolate bars. C) The machine is calibrated to produce chocolate bars with a mean weight of $\mu_0 = 100$ grams. Based on your confidence interval, what conclusion can be drawn about the calibration of the machine?

a. The machine is likely calibrated correctly, as μ_0 is within the interval.

b. The machine is likely not calibrated correctly, as μ_0 is outside the interval.

c. The machine is likely calibrated correctly, as the sample mean is close to μ_0.

d. The machine is likely not calibrated correctly, as the sample standard deviation is too large.

32 [Intermediate] • Target: The Central Limit Theorem

A sample of 5 numbers has a mean of 12 and a standard deviation of 3. If every number in the sample is multiplied by 2, what are the new mean and standard deviation?

a. Mean = 24, SD = 3

b. Mean = 24, SD = 6

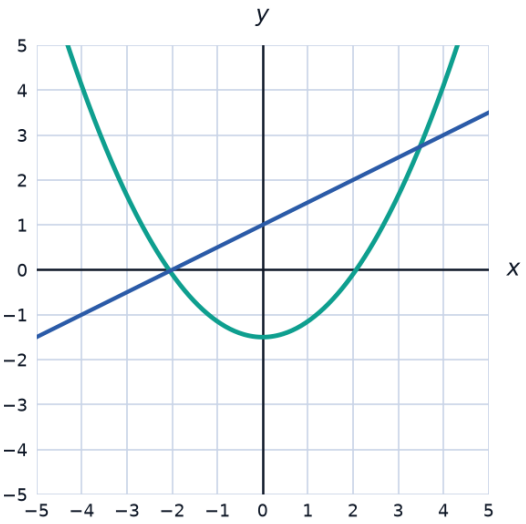
c. Mean = 12, SD = 6

d. Mean = 24, SD = 9

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33 [Intermediate] • Target: Applying the Central Limit Theorem

For a sample of size $n = 30$ (so $df = 29$), find the critical value t^* for a 99% confidence interval for the mean.



a. 3.256

b. 2.756

c. 3.056

d. 1.956

34 [Intermediate] • Target: Sampling distribution of a sample proportion

Increasing sample size n primarily:

a. Increases bias

b. Reduces variability

c. Increases variability

d. Has no effect on confidence intervals

35 [Intermediate] • Target: Comparing sampling variability across sample sizes

A randomised block design blocks subjects by gender into 3 blocks of 57 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 171

b. 156

c. 196

d. 186

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36 [Intermediate] • Target: Applying the sampling distribution of the sample mean

A randomised block design blocks subjects by fitness level into 3 blocks of 20 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 45

b. 85

c. 50

d. 60

37 [Intermediate] • Target: Sample size and the CLT threshold

A randomised block design blocks subjects by region into 2 blocks of 42 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 59

b. 84

c. 74

d. 109

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Standard error vs. standard deviation • Cross-Lesson: 3.01 Probability Rules & Conditional Probability

A randomised block design blocks subjects by prior experience into 3 blocks of 29 subjects each, with random assignment to treatments within each block. What is the total sample size?

a. 77

b. 97

c. 87

d. 72

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39 [Challenging] • Target: What is a sampling distribution? • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A political poll surveyed 1200 likely voters. The results indicated that 55% of respondents plan to vote for Candidate A, with a margin of error of ± 3 percentage points. If the poll is adjusted to include only likely voters who are registered as "strong supporters" (who constitute 60% of the overall surveyed population), and assuming the proportion of "strong supporters" who favor Candidate A remains the same as the overall proportion, what is the most accurate statement regarding the new confidence interval for the proportion of strong supporters who favor Candidate A? Let P be the proportion of all likely voters who plan to vote for Candidate A, and let n be the sample size. The margin of error for a proportion is typically calculated as $z\sqrt{\frac{P(1-P)}{n}}$, where z is the critical value for the desired confidence level. For the initial poll: $P = 0.55$ $n = 1200$ Margin of Error = 0.03 This implies a confidence interval for P of $[0.52, 0.58]$. Now, consider the subset of "strong supporters," which represents 60% of the population. Let P_{ss} be the proportion of strong supporters who favor Candidate A. We are given that the proportion of "strong supporters" who favor Candidate A is assumed to be the same as the overall proportion, so $P_{ss} \approx P = 0.55$. The sample size for this subset, assuming it's representative, would be $n_{ss} = 0.60 \times n = 0.60 \times 1200 = 720$. The new margin of error for this subgroup would be $ME_{ss} = z\sqrt{\frac{P_{ss}(1-P_{ss})}{n_{ss}}}$. Since $n_{ss} < n$, the margin of error will increase (making the interval wider) if the proportion P_{ss} were significantly different, or it would decrease if n_{ss} were larger. However, if the proportion is the same, a smaller sample size will lead to a larger margin of error.

a. The new confidence interval for strong supporters will be wider than the original interval.

b. The new confidence interval for strong supporters will be narrower than the original interval.

c. The new confidence interval for strong supporters will be exactly the same as the original interval.

d. It is impossible to determine if the new confidence interval is wider or narrower without knowing the z-score.

40 [Challenging] • Target: Mean and standard deviation of the sampling distribution • Cross-Lesson: 3.03 The Normal Distribution

A data set of 100 observations has a mean of 50 and standard deviation 10. According to Chebyshev's inequality, at least what percentage of the data must lie between 30 and 70?

a. 50%

b. 68%

c. 75%

d. 95%

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41 [Challenging] • Target: The Central Limit Theorem • Cross-Lesson: 4.01 Confidence Intervals & Hypothesis Testing

A researcher wants to estimate a population proportion with a margin of error of at most $\pm 2\%$ at a 95% confidence level. An earlier study with $n = 400$ had a margin of error of $\pm 5\%$. Approximately what sample size n is required?

a. 1,000

b. 1,600

c. 2,500

d. 4,000

42 [Challenging] • Target: Applying the Central Limit Theorem • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

A data set X has mean μ and standard deviation σ. A new data set Y is formed by the transformation $y_i = 3x_i - 5$ for each value in X. What are the mean and standard deviation of Y?

a. Mean = $3\mu - 5$, SD = $3\sigma - 5$

b. Mean = $3\mu - 5$, SD = 3σ

c. Mean = 3μ, SD = 3σ

d. Mean = $\mu - 5$, SD = σ

43 [Challenging] • Target: Sampling distribution of a sample proportion • Cross-Lesson: 1.01 Measures of Center & Spread

A student takes 4 tests and has a mean score of 85 with standard deviation 0. What were the student's individual test scores?

a. 80, 85, 85, 90

b. All four scores were 85

c. 75, 85, 85, 95

d. 82, 84, 86, 88

44 [Challenging] • Target: Comparing sampling variability across sample sizes • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

In estimating a population proportion, what is the effect on the margin of error of increasing the sample size (with the same confidence level), assuming all else is held constant?

a. The margin of error tends to increase

b. The margin of error tends to decrease

c. The margin of error stays exactly the same

d. The margin of error becomes zero

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45 **[Challenging]** • **Target: Applying the sampling distribution of the sample mean** • **Cross-Lesson: 1.03 Describing Distributions & Linear Regression**

In estimating a population proportion, what is the effect on the margin of error of decreasing the sample size (with the same confidence level), assuming all else is held constant?

a. The margin of error tends to increase

b. The margin of error tends to decrease

c. The margin of error stays exactly the same

d. The margin of error becomes zero

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